

1 Power of a point - the basics

Let Γ be a circle with center O and radius r . Then the power of a point P with respect to Γ is defined as

$$\Pi_{\Gamma}(P) = PO^2 - r^2.$$

Other ways to describe this:

1. If P is outside the circle Γ : if T is a point on Γ such that PT is tangent to Γ , then $\Pi_{\Gamma}(P) = PT^2$.
2. If P is inside the circle Γ , but not equal to O : if A is a point on Γ such that $AP \perp OP$, then $\Pi_{\Gamma}(P) = -AP^2$.
3. If $P = O$: $\Pi_{\Gamma}(O) = -r^2$.

Some basic useful results:

1. Two intersecting circles $\Gamma_1 = (O_1, r_1)$ and $\Gamma_2 = (O_2, r_2)$ are orthogonal iff $\Pi_{\Gamma_1}(O_2) = r_2^2$.
2. If $\Gamma = (O, r)$ is a circle and A' is the image of A under inversion in Γ , then

$$\begin{aligned} \Pi_{\Gamma}(A) + \Pi_{\Gamma}(A') &= AA'^2, \\ \Pi_{\Gamma}(A) \cdot \Pi_{\Gamma}(A') &= -r^2 \cdot AA'^2, \\ \frac{1}{\Pi_{\Gamma}(A)} + \frac{1}{\Pi_{\Gamma}(A')} &= -\frac{1}{r^2}. \end{aligned}$$

Introductory Problems:

1. Show that the locus of a point whose sum of powers with respect to two given circles is constant is a circle with center the midpoint of the segment connecting the two centers of the circles.
2. Show that the locus of a point whose sum of powers with respect to three given circles is constant is a circle with center the centroid of the triangle formed by the three centers of the circles. What happens when the triangle is degenerate?
3. Let H be the orthocenter of $\triangle ABC$, let Γ be its circumcircle and let Γ^* be its nine-point circle.
 - (a) Show that $\Pi_{\Gamma}(H) = 4\Pi_{\Gamma^*}(H)$.
 - (b) Show that $\Pi_{\Gamma^*}(A) + \Pi_{\Gamma^*}(B) + \Pi_{\Gamma^*}(C) = \frac{1}{4}(AB^2 + BC^2 + AC^2)$
4. Let $ABCD$ be a rectangle inscribed in a circle $\Gamma = (O, r)$. Let P be a point in the interior of the rectangle, let PX , PX' , PY , and PY' be the segments perpendicular to the sides AB , CD , AD , and BC , respectively, (With the points X, X', Y, Y' on these sides). Prove that $PX \cdot PX' + PY \cdot PY' = -\Pi_{\Gamma}(P)$.
5. Let the lines joining the vertices A , B , and C of a triangle ABC to a point S meet the respective opposite sides in L , M , and N . Furthermore, let the circle (LMN) meet these sides again in L' , M' , and N' , respectively; show that the lines AL' , BM' , and CN' are concurrent.

Harder Problems:

1. Let P be a point inside a circle such that there exist three chords through P of equal length. Prove that P is the center of the circle.
2. For a point P inside the angle XOY , find A on OX and B on OY such that P is on AB and the product $AP \cdot BP$ is minimal.
3. In triangle ABC let O , I and H be the circumcenter, the incenter, and the orthocenter, respectively. Compute the lengths of the segments OI and OH in terms of the circumradius, the inradius, and the angles of the triangle ABC .
4. Let Γ_1 and Γ_2 be concentric circles, with Γ_2 in the interior of Γ_1 . From a point A on Γ_1 draw the tangent AB to Γ_2 (with B on Γ_2). Let C be the second point of intersection of AB with Γ_1 , and let D be the midpoint of AB . A line passing through A intersects Γ_2 at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB . Find, with proof, the ratio AM/MC .
5. Let ABC be an acute triangle. The points M and N are taken on the sides AB and AC , respectively. The circles with diameters BN and CM intersect at points P and Q . Prove that P , Q , and the orthocenter H are collinear.
6. Let $ABCD$ be a convex quadrilateral inscribed in a semicircle σ of diameter AB . The lines AC and BD intersect at E and the lines AD and BC at F . The line EF intersects the semicircle σ at G and the line AB at H . Prove that E is the midpoint of the line segment GH iff G is the midpoint of the line segment FH .

2 Radical axes

The *radical axis* of two circles $\Gamma_1 = (O_1, r_1)$ and $\Gamma_2 = (O_2, r_2)$ such that $O_1 \neq O_2$ is the locus of all points that have equal powers with respect to the two circles.

Some basic properties:

1. The radical axis is a line that is perpendicular to the line O_1O_2 .
2. If the two circles intersect it is the line through the two intersection points.
3. If the two circles are tangent in point T , the radical axis is the common tangent line through T .
4. If the centers O_1 , O_2 and O_3 of three circles are not collinear, their radical axes intersect in a common point, the *radical centre*.
5. If Γ_3 has its centre O_3 on the radical axis of Γ_1 and Γ_2 and intersects both Γ_1 and Γ_2 , then Γ_3 is orthogonal to Γ_1 if and only if it is orthogonal to Γ_2 . Furthermore, if Γ_3 is orthogonal to both Γ_1 and Γ_2 , its centre lies on the radical axis of Γ_1 and Γ_2 .

When the two circles do not intersect, here are some ways to find the radical axis:

- **Algebraically** If R is the intersection point of the radical axis with O_1O_2 , then we can determine its position based on the following two equations:

$$O_1R^2 - r_1^2 = O_2R^2 - r_2^2 \text{ and } O_1R + O_2R = O_1O_2$$

- **Construction, I** For any circle $\Gamma_3 = (O_3, r_3)$, with O_3 not on O_1O_2 , that intersects Γ_1 and Γ_2 , you can find the radical centre, which lies on the radical axis of Γ_1 and Γ_2 . If you take the circle with center the midpoint of O_1O_2 , how does it determine the radical axis?
- **Construction, II** The midpoint of any common tangent line lies on the radical axis.

Introductory Problems

1. Let P be a point on the radical axis of two circles Γ_1 and Γ_2 . Show that the polars of P with respect to Γ_1 and Γ_2 intersect on the radical axis.
2. Let ABC be a triangle and consider a point P in its interior. Suppose that BC is tangent to the circumcircles of triangles ABP and ACP . Prove that ray AP bisects the segment BC .
3. Let P be a point on the radical axis of two circles Γ_1 and Γ_2 . Let ℓ_1 and ℓ_2 be secants of Γ_1 and Γ_2 respectively that pass through P . Then the intersection points of ℓ_1 with Γ_1 together with the intersection points of ℓ_2 with Γ_2 are concyclic.
4. Show that the orthic axis of a triangle is the radical axis of the circumcircle and the nine-point circle of the triangle.
5. Show that the four tritangent circles of a triangle ABC taken in pairs have for their radical axes the bisectors of the angles of the medial triangle of ABC .
6. Show that the radical axis of the two circles having for diameters the diagonals AC , BD of a trapezoid $ABCD$ passes through the point of intersection E of the nonparallel sides BC and AD .
7. Show that the locus of the centers of circles which bisect two given circles $\Gamma_1 = (O_1, r_1)$ and $\Gamma_2 = (O_2, r_2)$ is a straight line perpendicular to O_1O_2 . This line is sometimes called the *anti-radical axis* of the two circles. Show furthermore that the radical axis and the anti-radical axis are symmetric with respect to the midpoint of O_1O_2 .

Harder Problems

1. Let ABC be a triangle and let A' , B' and C' be points on sides BC , CA , and AB , respectively. Denote by M the point of intersection of circles (ABA') and $(A'B'C')$ other than A' , and by N the point of intersection of circles ABB' and $A'B'C'$ other than B' . Similarly, one defines points P , Q and R , S respectively. Prove that:

- (a) At least one of the following situations occurs:
- the triples of lines
 $(AB, A'M, B'N), (BC, B'P, C'Q)$ and $(CA, C'R, A'S)$
are concurrent at C'', A'' and B'' , respectively;
 - $A'M$ and $B'N$ are parallel to AB , or $B'P$ and $C'Q$ are parallel to BC , or $C'R$ and $A'S$ are parallel to CA .
- (b) In the case where (i) occurs, the points A'', B'' and C'' are collinear.
- Among the points A, B, C , and D , no three are collinear. The lines AB and CD intersect at E , and BC and DA intersect at F . Prove that either the circles with diameters AC, BD , and EF pass through a common point, or no two of them have any common points.
 - Let ABC be a triangle and let D and E be points on the sides AB and AC , respectively, such that DE is parallel to BC . Let P be any point interior to the triangle ADE and let F and G be the intersections of DE with the lines BP and CP , respectively. Let Q be the second intersection point of the circumcircles of triangles PDG and PFE . Prove that the points A, P and Q are collinear.
 - Let A, B, C and D be four distinct points on a line, in this order. The circles with diameters AC and BD intersect in the points X and Y . The line XY meets BC at Z . Let P be a point on XY other than Z . The line CP intersects the circle with diameter BC at C and M , and the line BP intersects the circle with diameter BD at B and N . Prove that the lines AM, DN , and XY are concurrent.
 - Consider the semicircle of center O and diameter AB . A line intersects AB at M and the semicircle at C and D in such a way that $MB < MA$ and $MD < MC$. The circumcircles of triangles AOC and DOB intersect a second time at K . Show that MK and KO are perpendicular.

Competition Problems

- (USAMO 2009) Given circles ω_1 and ω_2 intersecting at points X and Y , let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S . Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY .
- (Shortlist 1994) A circle ω is tangent to two parallel lines ℓ_1 and ℓ_2 . A second circle ω_1 is tangent to ℓ_1 at A and to ω externally at C . A third circle ω_2 is tangent to ℓ_2 at B , to ω externally at D , and to ω_1 externally at E . AD intersects BC at Q . Prove that Q is the circumcenter of triangle CDE .
- (IMO 1985) A circle with center O passes through points A and C and intersects the sides AB and BC of the triangle ABC at points K and N , respectively. The

- circumscribed circles of the triangles ABC and KBN intersect at two distinct points B and M . Prove that $\angle OMB = 90^\circ$.
4. (Shortlist 1992) Circles Γ , Γ_1 , Γ_2 are related to each other as follows: circles Γ_1 and Γ_2 are externally tangent to one another at a point W and both these circles are internally tangent to the circle Γ . Points A , B , and C are located on the circle Γ as follows: Line BC is a direct common tangent to the pair of circles Γ_1 and Γ_2 , and the line WA is the transverse common tangent at W to Γ_1 and Γ_2 , with W and A lying on the same side of the line BC . Prove that W is the incenter of the triangle ABC .
5. (Iran TST 2011) In acute triangle ABC , $\angle B$ is greater than $\angle C$. Let M be the midpoint of BC and let E and F be the feet of the altitudes from B and C , respectively. Let K and L be the midpoints of ME and MF respectively and let T be on line KL such that $AT \parallel BC$. Prove $AT = TM$.